TURNING CHURN DATA INTO
RETAINED REVENUE

A PREDICTIVE-TARGETING PROPOSAL FOR COMPANY A (WIRELESS TELECOM)

CREATED BY: ISHRAQ KAMAL ADIB

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EXECUTIVE SUMMARY

SPEND RETENTION BUDGET WHERE IT PAYS OFF

RECOMMENDATION: SCORE EVERY CUSTOMER'S CHURN RISK MONTHLY; TARGET ONLY THE TOP 10% WITH A MATCHED RETENTION OFFER, INSTEAD OF SPREADING BUDGET ACROSS THE WHOLE BASE.

0.691

MODEL RANKING QUALITY (ROC-AUC), XGBOOST

+$333K

NET VALUE OF ONE TARGETING CYCLE (CONSERVATIVE CASE)

~5.6×

VALUE VS. SPENDING THE SAME BUDGET UNTARGETED

TOP-DECILE TARGETING CAPTURES ~16% OF NEXT-MONTH CHURNERS; ~30% AT TOP 20%.

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MARKET CONTEXT

WHY RETENTION, AND WHY NOW

1

SATURATED MARKET

US WIRELESS IS HIGHLY PENETRATED; POSTPAID MONTHLY CHURN RUNS ~0.9–1.3%. GROWTH NOW COMES FROM TAKING SHARE, NOT NEW DEMAND. [1][2][3]

2

RETENTION IS FAR CHEAPER

ACQUIRING A CUSTOMER COSTS ROUGHLY 5–25× MORE THAN RETAINING ONE; A 5% RETENTION GAIN CAN LIFT PROFITS 25–95% (BAIN). [4]

3

RECURRING REVENUE COMPOUNDS

VALUE = ARPU × TENURE (ARPU ≈ $58/MO HERE). SMALL CHURN REDUCTIONS COMPOUND INTO LARGE LIFETIME-VALUE GAINS.

PROBLEM

THE PROBLEM: CHURN THE CLIENT CAN'T SEE COMING

BUDGET IS SPREAD UNIFORMLY.

EVERY CUSTOMER GETS THE SAME (LITTLE) ATTENTION, SO SPEND IS WASTED ON THE ~80% WHO WOULD NEVER LEAVE.

THERE IS NO PER-CUSTOMER RISK RANKING.

THE CLIENT CANNOT TELL, IN ADVANCE, WHICH CUSTOMERS ARE ABOUT TO CHURN — SO INTERVENTION IS REACTIVE, AFTER THE CANCEL REQUEST.

DATA CAVEAT WE CORRECTED FOR

THE ANALYSIS SAMPLE IS 49.6% CHURNERS — OVERSAMPLED BY DESIGN, NOT THE REAL RATE. TRUE MONTHLY CHURN IS ~1– 2% [1][2]. EVERY REVENUE FIGURE IN THIS DECK USES THE CORRECTED BASE RATE, NOT THE INFLATED SAMPLE RATE.

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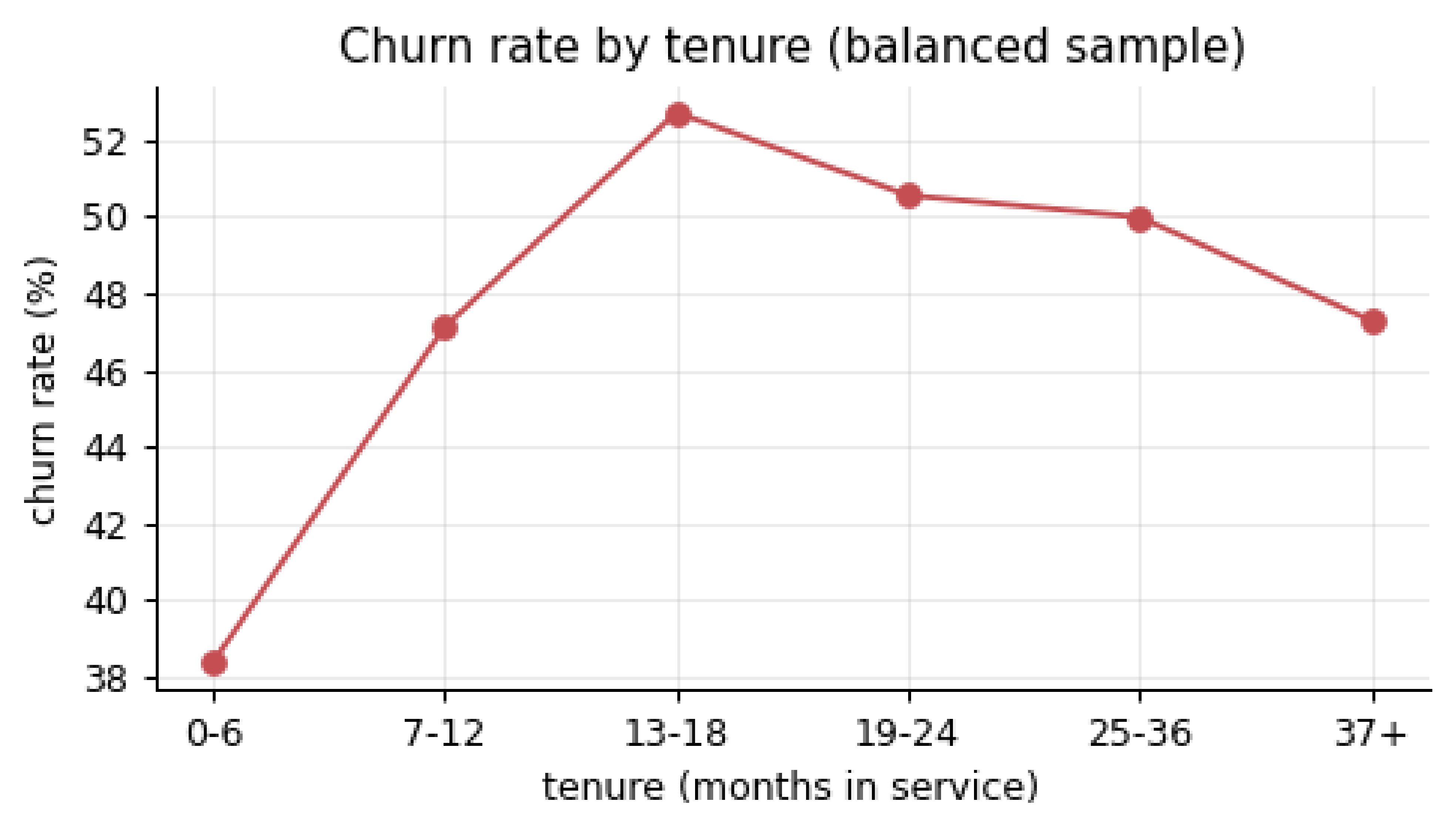
DATA & APPROACH

THE DATASET

- 100,000 CUSTOMERS, ~100 FEATURES
- USAGE, BILLING, HANDSET, TENURE (FIRST-PARTY) + APPENDED DEMOGRAPHICS
- TARGET: CHURN WITHIN 31–60 DAYS OF THE OBSERVATION DATE
- TWO TABLES JOINED 1:1 ON CUSTOMER ID

1. UNDERSTAND THE MARKET & DATA
2. FORM A CHURN HYPOTHESIS
3. BUILD & COMPARE ML MODELS
4. TRANSLATE MODEL INTO A PROPOSAL
5. VERIFY IMPACT & LIMITATIONS

WHAT THE DATA SHOWS



- RISK IS LOWEST IN THE FIRST 6 MONTHS, THEN CLIMBS AND PLATEAUS NEAR THE CONTRACT-RENEWAL WINDOW.
- APPENDED DEMOGRAPHICS ARE HEAVILY MISSING (CAR OWNERSHIP 49%, DWELLING 38%, INCOME 25%) AND MISSING-NOT-AT-RANDOM.
- THE MODEL LEANS ON FIRST-PARTY USAGE & BILLING SIGNAL, WHICH IS COMPLETE AND ACTIONABLE.

HYPOTHESIS

THREE ACTIONABLE CHURN LEVERS

HANDSET & CONTRACT LIFECYCLE

OLDER EQUIPMENT AND APPROACHING CONTRACT END RAISE RISK.

ACTION: PROACTIVE UPGRADE / RE-CONTRACT OFFERS.

USAGE DISENGAGEMENT

FALLING MINUTES AND REVENUE VS. A CUSTOMER'S OWN BASELINE PRECEDE CHURN

ACTION: WIN-BACK WHEN USAGE DROPS.

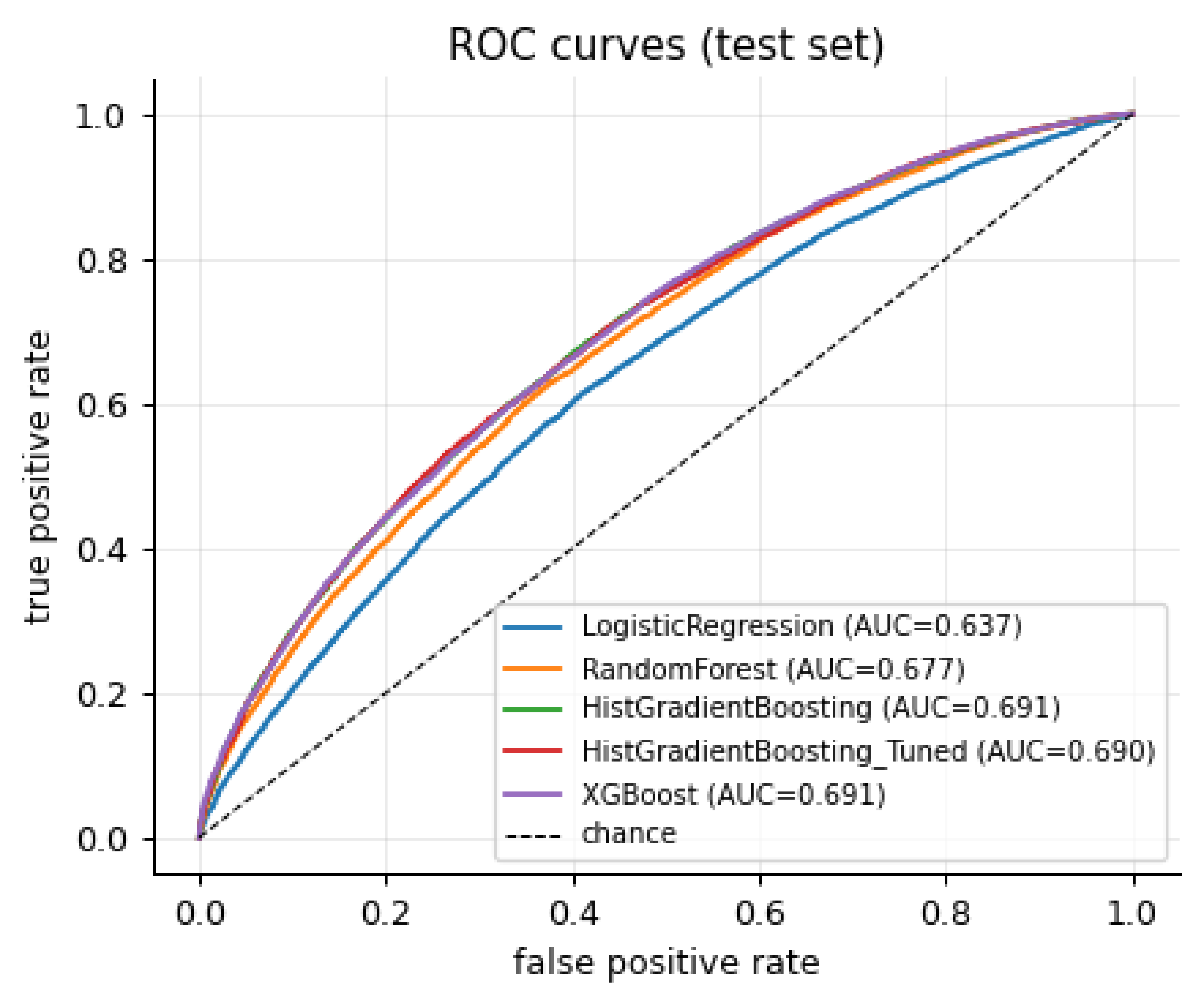
SERVICE FRICTION

DROPPED/BLOCKED CALLS AND HEAVY CUSTOMER-CARE CONTACT CORRELATE WITH LEAVING.

ACTION: TARGETED SERVICE RECOVERY.

MODELING

A MODEL THAT RANKS CUSTOMERS BY RISK



SELECTED MODEL

XGBOOST

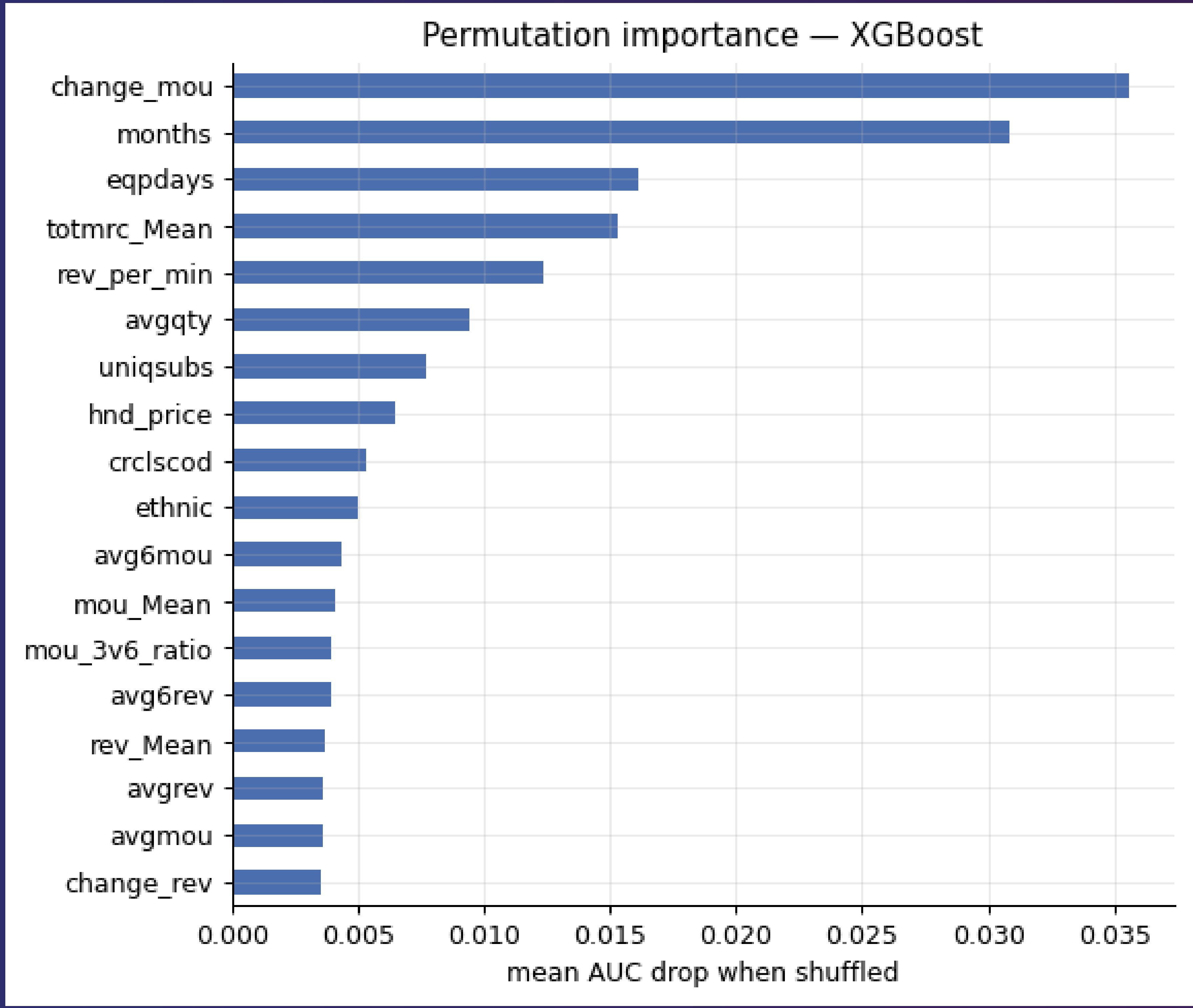
ROC-AUC 0.691

TOP-DECILE LIFT 1.60×

THE TOP THREE BOOSTED MODELS ARE STATISTICALLY TIED AT AUC ≈ 0.69; XGBOOST EDGES AHEAD ON TEST AUC AND LIFT. AUC ≈ 0.69 IS REALISTIC FOR THIS LOW-SIGNAL DATA — VALUE IS IN RELIABLE RANKING, NOT PERFECT CLASSIFICATION. TEXT

ACCURACY NOT USED AS THE DECISION METRIC (BALANCED SAMPLE); RANKING METRICS REPORTED INSTEAD.

WHAT DRIVES CHURN AND WHAT WE CAN DO



MAPPED TO LEVERS

DISENGAGEMENT: CHANGE IN MINUTES (CHANGE_MOU), 3-VS-6-MO USAGE RATIO

LIFECYCLE: TENURE (MONTHS), HANDSET AGE (EQPDAYS), HANDSET PRICE

ECONOMICS: RECURRING CHARGE (TOTMRC), REVENUE/MINUTE, UNIQUE SUBS

TOP PREDICTORS FROM THE XGBOOST MODEL. TENURE PREDICTS STRONGLY BUT IS USED FOR PRIORITISATION, NOT INTERVENTION

THE PROPOSAL: TARGETED MONTHLY RETENTION

HOW IT RUNS

1. SCORE THE WHOLE BASE FOR CHURN RISK EACH MONTH

2.ROUTE THE TOP 10% INTO A RETENTION PROGRAMME

3.MATCH THE OFFER TO THE CUSTOMER'S RISK LEVER (UPGRADE /
WIN-BACK / SERVICE FIX)

4.HOLD OUT A RANDOM CONTROL GROUP TO MEASURE TRUE EFFECT

+$333K

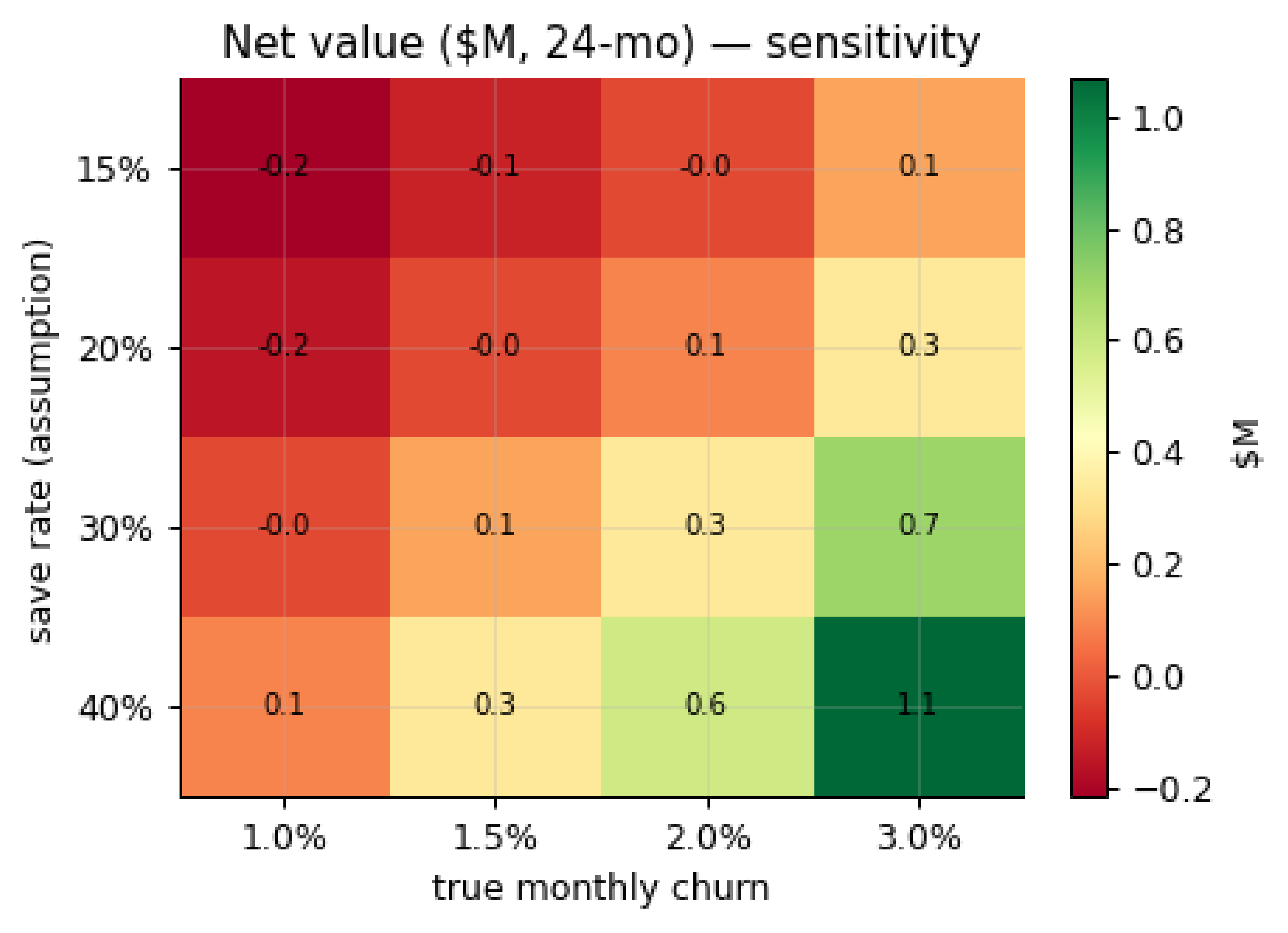
NET VALUE, ONE TARGETING CYCLE (CONSERVATIVE CASE)

+$274K

ATTRIBUTABLE TO THE MODEL — VS ONLY $59K FOR T
HE SAME BUDGET SPENT AT RANDOM

ROI 0.8× (RANGE 0.7×–2.2× ACROSS PLAUSIBLE MARGIN/HORIZON)

IMPACT HOLDS ACROSS THE ASSUMPTIONS



- NET VALUE (24-MO) AS TRUE CHURN AND THE SAVE RATE VARY — THE TWO ASSUMPTIONS THE CASE IS MOST SENSITIVE TO.
- GREEN = POSITIVE. THE PROPOSAL PAYS OFF ACROSS MOST OF THE PLAUSIBLE REGION.
- IT ONLY TURNS NEGATIVE IN THE PESSIMISTIC CORNER (LOW CHURN AND LOW SAVE RATE).
- THE HEADLINE USES A CONSERVATIVE MID-POINT, NOT A BEST CASE.

SAVE RATE AND COST-PER-CONTACT ARE ASSUMPTIONS (NO PUBLIC BENCHMARK); CHURN AND MARGIN ARE SOURCED

VALIDATE ON A REAL BASE RATE

RE-MEASURE LIFT ON A REPRESENTATIVE (NON-OVERSAMPLED) SAMPLE BEFORE COMMITTING TO DOLLAR FIGURES.

PROVE CAUSALITY WITH AN A/B HOLDOUT

CORRELATION ≠ SAVED CUSTOMER. A RANDOMISED CONTROL MEASURES THE TRUE INCREMENTAL RETENTION.

RECALIBRATE PROBABILITIES

RESCALE SCORES TO THE TRUE ~2% PRIOR BEFORE TREATING THEM AS PROBABILITIES.

RESPECT THE SIGNAL CEILING

AUC ≈ 0.69 IS GENUINE FOR THIS DATA;
THE TOP BOOSTED MODELS TIE,
SO THE CHOICE IS ROBUST.
VALUE = RANKING + CHEAP INTERVENTION.

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